SmartUSBHub Pro 4CH USB2.0

Technical Specifications

Applies to: HBP_USB2_4CH / HBP_USB2_4CH_PSU

Released v1.0

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Document item	Details
Applicable product	SmartUSBHub Pro 4CH USB2.0
Order codes	HBP_USB2_4CH, HBP_USB2_4CH_PSU
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1 Product Overview

SmartUSBHub Pro 4CH USB2.0 is a four-port programmable USB 2.0 High-Speed hub. It independently controls VBUS and USB 2.0 D+ / D- on every downstream port and measures each channel's VBUS voltage and output current. It is designed for R&D debugging, automated testing, production validation and remote USB device management.

1.1 Model Differences

Variant	Order code	Voltage/current measurement	Included power supply
Standard	HBP_USB2_4CH	Yes	Not included
Power-adapter bundle	HBP_USB2_4CH_PSU	Yes	5 V / 2 A

1.2 Key Features

- Four USB-A downstream ports with independent VBUS and D+ / D- control.
- USB 2.0 High-Speed link rate up to 480 Mbps; compatible with Full-Speed, Low-Speed and USB 1.1 devices.
- MTT architecture with an independent transaction translator for each downstream port.
- Independent VBUS voltage and output-current measurement on every channel.
- USB BC 1.2 CDP support with up to 1.5 A charging current per port; QC, USB PD and other fast-charging protocols are not supported.
- Normal mode, interlock mode, configurable power-on defaults and state restoration after power loss.
- Four local channel-control buttons.

2 System Architecture

2.1 System Block Diagram

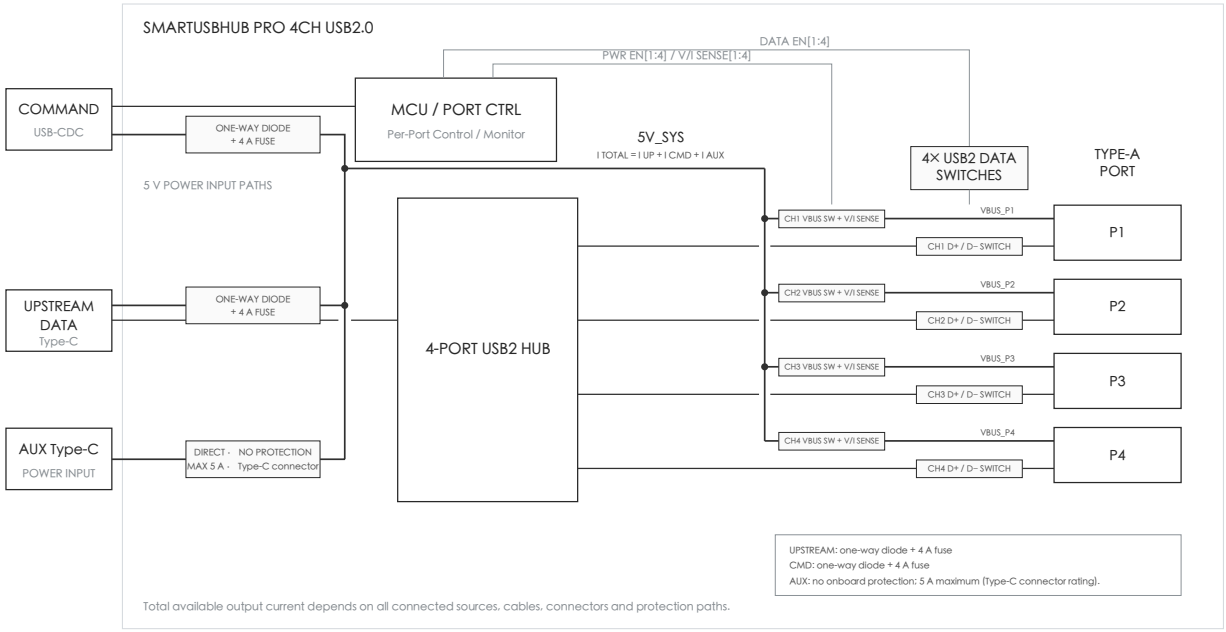


Figure 2-1 SmartUSBHub Pro 4CH USB2.0 system block diagram

The DATA port carries USB traffic between downstream devices and the host. The CMD port enumerates as a USB CDC serial interface and receives control commands. DATA, CMD and AUX POWER can all feed the internal 5 V rail.

2.2 Interfaces and Indicators

Interface or component	Quantity	Function
USB-C DATA	1	Connects to the USB host and carries downstream-device data
USB-C CMD	1	Connects to the control host and provides the USB CDC command interface
USB-C AUX POWER	1	Accepts a protected external 5 V DC supply
USB-A downstream ports	4	Connect controlled USB devices
Channel buttons	4	Control the corresponding channels
Status indicator	1	Indicates device operation and command processing
Channel indicators	4	Indicate the VBUS state of the corresponding channels

3 Functional Specifications

3.1 USB Data Path

Parameter	Specification
USB standard	USB 2.0 High-Speed
Maximum link rate	480 Mbps
Compatible speeds	Full-Speed 12 Mbps, Low-Speed 1.5 Mbps, USB 1.1
Upstream data port	1 × USB-C
Downstream ports	4 × USB-A
Hub architecture	MTT, independent TT for every downstream port

3.2 Channel Control

Parameter	Specification
Channel numbers	CH1–CH4
VBUS control	Independent on every downstream port
D+ / D- control	Independent on every downstream port
Power/data relationship	USB data can be disconnected while VBUS remains powered
Batch control	Supported
Normal mode	Supported
Interlock mode	Supported; only one channel can be enabled at a time
Power-on default state	Configurable
State restoration after power loss	Configurable

3.3 Charging and Output Path

Parameter	Specification
Charging protocol	USB BC 1.2 CDP
BC 1.2 charging current	Up to 1.5 A per port
Per-channel output-path capability	Up to 4 A; this is not a guaranteed charging current
Other fast-charging protocols	QC, USB PD and other fast-charging protocols are not supported

3.4 Voltage and Current Measurement

Measurement	Range	Effective resolution	Measurement accuracy
VBUS voltage	0–5.5 V/channel	10 mV (typical)	Status monitoring only; not a precision measurement
Current	Approx. 30 mA–4.0 A/channel	10 mA (typical)	Uncalibrated; trend monitoring only

Voltage and current data are reported in units of 1 mV and 1 mA, respectively; the reporting unit does not represent effective measurement resolution. Current below approximately 30 mA may be reported as zero. Measurements are affected by the voltage reference, shunt resistor, amplifier error, temperature, and component variation. They are not intended for metrology or precision power measurements; use a calibrated external instrument when an accurate determination is required.

4 Electrical Specifications



WARNING: input overvoltage can cause overheating or fire

Only connect a regulated 5 V DC supply to any power input. No input may exceed 5.5 V. Never use a PD trigger, boost cable or other method to apply 9 V, 12 V, 15 V or 20 V to AUX POWER.

Parameter	Min.	Typ.	Max.	Unit	Notes
5 V input voltage	4.75	5.00	5.25	V	Recommended operating range
Absolute maximum input	0	-	5.5	V	Applies to all power inputs
Downstream-port VBUS	4.5	5.0	5.5	V	Without overload
Total input current	-	-	<10	A	Sum of DATA, CMD and AUX POWER currents

Parameter	Min.	Typ.	Max.	Unit	Notes
Per-channel output-path capability	-	-	4	A	Does not represent charging-protocol current
Included adapter output	-	2	-	A	Shared by all four ports



CAUTION: size the power supply for the actual load

If four devices each draw the USB BC 1.2 CDP maximum of 1.5 A, the theoretical load is approximately 5 V / 6 A. The combined capacity of the three power inputs must cover the actual load, and total input current must remain below 10 A.

5 Protection and Safety Limits

Item	Specification
DATA and CMD power paths	Reverse-current, overvoltage and overcurrent protection
Downstream ports	Resettable fuse on every port
AUX POWER	No independent onboard overcurrent protection; the external supply must provide current limiting and short-circuit protection
Input voltage	Regulated 5 V DC only; 5.5 V absolute maximum

- The 4 A per-channel rating is the output-path capability, not a guaranteed charging current.
- USB BC 1.2 CDP charging current is limited to 1.5 A per port.
- Available output current depends on the supply, cables, connectors, ambient temperature and cooling.
- Turning off VBUS or D+ / D- is equivalent to unplugging a device. Stop file writes, firmware updates and other non-interruptible operations first.

6 Mechanical and Environmental Specifications

Parameter	Specification
Dimensions	106 mm × 46 mm × 18 mm
Device weight	52 g
Enclosure material	PMMA
Mounting	Supports stacked mounting with standoffs
Operating temperature	0–50 ° C, non-condensing
Storage temperature	-10–85 ° C, non-condensing

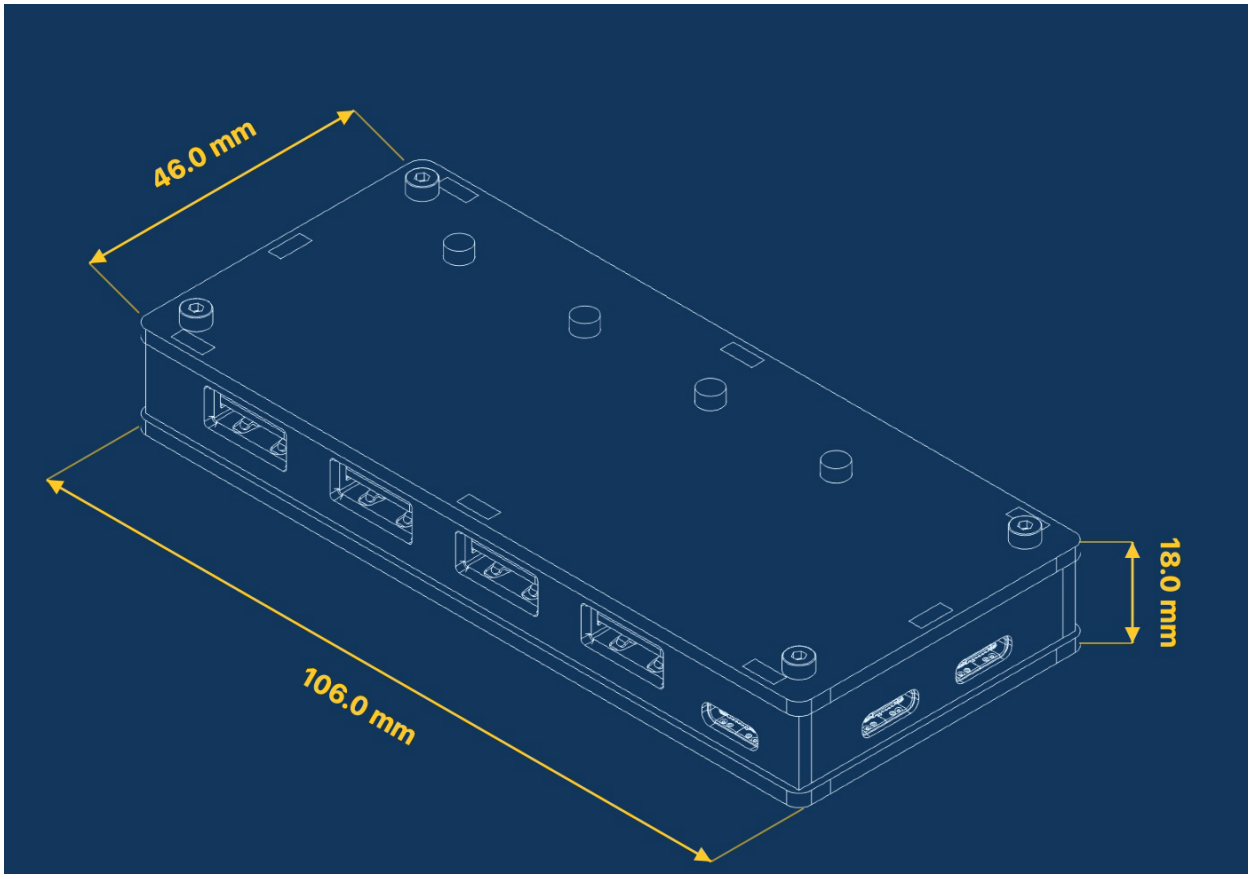


Figure 6-1 Product outline and dimensions

7 Related Documentation

Document	Link
SmartUSBHub Pro 4CH USB2.0 User Manual	View manual
SmartUSBHub communication protocol	View documentation
SmartUSBHub documentation center	View documentation
SmartUSBHub Python library guide	View documentation